

Phase 6 - Project Testing

PocketSmart AI: Your Smart Budget & Recommendation Assistant

Date: 29 September 2026

1. Testing Overview

Testing validates PocketSmart AI across the three planners, authentication, AI integration, sessions, recommendation display and fallback behaviour.

2. Testing Objectives

- Verify planner input handling.
- Verify recommendation generation.
- Check budget adherence.
- Evaluate Gemini response quality.
- Check platform/source accuracy.
- Validate authentication and sessions.
- Test edge cases and invalid inputs.
- Verify fallback/default recommendations.

3. Test Approach

Testing progresses from individual forms/routes to complete workflows: input → FastAPI route → recommendation service → Gemini/catalog logic → result display.

Project Testing - Authentication and Home

4. Authentication Tests

ID	Scenario	Expected
AUTH-01	Valid registration	Account created
AUTH-02	Invalid/duplicate registration	Validation/error
AUTH-03	Valid login	Authenticated
AUTH-04	Invalid login	Rejected
AUTH-05	Logout	Session terminated
AUTH-06	Protected route without auth	Access restricted

5. Home Planner Tests

ID	Scenario	Expected
HOME-01	Valid budget/room/items	Recommendations
HOME-02	Missing input	Validation
HOME-03	Budget-constrained request	Budget context respected
HOME-04	AI unavailable	Fallback/default response

Project Testing - Party and Jewelry

6. Party Planner Tests

ID	Scenario	Expected
PARTY-01	Valid event/guests/budget	Recommendations
PARTY-02	Missing budget/guest	Validation
PARTY-03	Different event types	Context-specific output
PARTY-04	AI failure	Fallback/default response

7. Jewelry Planner Tests

ID	Scenario	Expected
JEW-01	Budget + occasion + style	Recommendations
JEW-02	No image	Text-based result
JEW-03	Valid outfit image	Image + text processing
JEW-04	Invalid image	Validation/error

8. Integration Checks

- Each frontend planner reaches its correct backend route.
- Authenticated user information is available to protected functionality.
- Recommendation details and history return expected user-specific information.

Project Testing - Quality and Reporting

9. Quality Checks

- UI/UX – forms, navigation, formatting and responsiveness.
- Reliability – AI failure and fallback handling.
- Security – authentication and environment-based credentials.
- Accuracy – budget adherence, contextual relevance and platform/source accuracy.
- Maintainability – modular route/service structure.

10. Optimization

- Refine prompts when output is unclear or exceeds budget.
- Improve validation for edge cases.
- Improve session handling.
- Improve recommendation formatting.
- Retest after significant changes.

11. Test Reporting

Actual executions should record Test ID, input/scenario, expected result, actual result, status, defect, correction and retest result. This document does not invent unverified pass/fail results.

12. Conclusion

Testing and optimization ensure consistent workflows across Home, Party and Jewelry and suitable fallback behaviour.